M. Omar Nawaz, PhD, FHEA

Lecturer in Climate Change Science

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Website	www.omarnawaz.com	**ORCID**	orcid.org/0000-0001-7706-7287

RESEARCH OVERVIEW

I am a research group leader and educator in air quality science working to understand how human activity and societal development influence the climate, exacerbate air quality, and harm public health. My programme integrates diverse and interdisciplinary perspectives to explore how big data and AI can inform environmental exposures, health impacts, and policymaking to improve population health and ameliorate global exposure inequalities. In service to my aims, I develop health impact and exposure assessment frameworks, drawing on high performance computing, modelling, machine learning, and data science.

PROFESSIONAL APPOINTMENTS

Cardiff University, Cardiff, Wales, UK

2025- Lecturer in Climate Change Science, Department of Earth and Environmental Sciences, School of Physical, Chemical, and Environmental Sciences

George Washington University, Washington, DC, USA

2023-2025 Postdoctoral Research Associate, Milken Institute School of Public Health
2024 Professorial Lecturer, Milken Institute School of Public Health
2023-2024 NRDC Health Science Policy Fellow, Milken Institute School of Public Health

University of Colorado Boulder, Boulder, CO, USA

2018-2023 PhD Research Assistant, Department of Mechanical Engineering

University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

2016-2018 MSc Research Assistant, Gillings School of Global Public Health

EDUCATION

2023 **PhD**, Air Quality, Department of Mechanical Engineering, CU Boulder
Research Supervisor: Professor Daven K. Henze
Thesis Title: *An adjoint sensitivity framework for public health: the sources of air pollution and their current and future impacts at the urban and national scale*
Topics: public health, exposure assessment, computer science, data science, atmospheric science

2018 **MSc**, Env. Engineering, Gillings School of Global Public Health, UNC Chapel Hill
Research Supervisor: Professor J. Jason West
Dissertation Title: *Benefits of reduced premature mortality from decreases in $PM_{2.5}$ and ozone in the United States from 1999 to 2015*

2017 **BSc**, Physics, Department of Physics and Astronomy, UNC Chapel Hill

2017 **BSc**, Applied Mathematics, Department of Mathematics, UNC Chapel Hill

RESEARCH FUNDING

Research Grants (funded role)

2026-2027 GW4, **Co-I**, *Mapping Displacement Futures: Towards a GW4 Centre on Displacement*. £2,506.

2026-2027 UKRI, **PI**, *Forward and adjoint modeling of transboundary secondary $PM_{2.5}$ in response to climate mitigation*. 75,000 Node-Hrs.

2025-2026 Wellcome Trust, **Institutional PI**, *Maximizing health benefits from reducing oil and gas emissions in Mexico.* £496,000*.

2025-2026 Supercomputing Wales, **PI**, *Estimating UK surface-level pollution from satellite data using machine learning and deterministic modelling*. 100,000 CPU-Hrs.

Research Grants (unfunded role)

2026-2029 Wellcome Trust, **Technical Panel Member**, *Comprehensive Assessment of Co-Benefits and Maladaptation of Solarisation in Thailand*. £1,339,120.

2024-2027 NASA, **Collaborator**, *Application of satellite observations in estimating NO_2 concentrations, mortality burdens, and inequities*. £112,000*.

Consultancy and Contract Research

2026 ICCT, **Consultant**, *Health impact assessment software development.* £13,000.
2026 ICCT, **Consultant**, *The health impacts of vehicle emissions in Mexico*. £2,000.
2026 ICCT, **Consultant**, *Health benefits of future Colombian transport*. £2,000.
2025-2026 ICCT, **Consultant**, *Health benefits of zero emission vehicles by 2050.* £12,000.
2024 ICCT, **Consultant**, *Global maritime health impacts and policy benefits*. £5,000.
2023-2025 ICCT, **Consultant**, *Study of global transportation health impacts*. £9,000*.

Under Review

2027-2030 UKRI, **PI**, *Mitigating and adapting to wildfire smoke at the wildland-urban interface under future climate and development scenarios*. £1,673,000*.

2027-2028 NERC, **PI**, *Quantifying the Cost of the Measurement Disparity of Geostationary Air Pollution Remote Sensing in Global South Cities*. £55,368.

Competitive Applications Shortlisted

2026 Wellcome Trust CDA, **PI**, *ACCEPT: Assessing Climate Change effects on Equity and Public health from Transboundary air pollution*. £2,110,000.

**Funding total converted to GBP from other currency and rounded to the nearest thousand*

HONOURS AND FELLOWSHIPS

2026 Fellow of the Higher Education Academy (FHEA)
2026 Cardiff University Enriching Student Life Awards Nominee
2026 GW4 Crucible 2026 Cohort
2024-2025 GeoCAFE Scholar
2023-2024 Natural Resources Defense Council Health Science Policy Fellowship
2018 Outstanding Mechanical Engineering Research Potential Fellowship
2018 1st place student poster award, UNC 5th Climate Change Symposium

PUBLICATIONS

Peer-Reviewed Publications (27)

2026 TOAR-II Health Collaborators, Tropospheric Ozone Assessment Report, Phase II: Changing Health Impacts of Global Ozone Exposure. *Philosophical Transactions of the Royal Society. Accepted.*

2026 Goldberg, D. L., Johnson, J., Shah, T., **Nawaz, M. O.**, Kerr, G. H., Chen, L., et al. (2026). Contrasting the TEMPO and TROPOMI views of tropospheric NO_2: Implications for diurnal variability and model evaluation. *Journal of Geophysical Research: Atmospheres*, 131, e2026JD046905. https://doi.org/10.1029/2026JD046905

2026 Huber, D.E., Kerr, G.H., **Nawaz, M.O.**, Runkel, S., Anenberg, S.C., Goldberg, D.L. TROPOMI NO_2 trends for urban and polluted areas globally from 2019 to 2024. *Atmos. Chem. Phys.,* **2026,** *26, 3783–3803,* https://doi.org/10.5194/acp-26-3783-2026.

2026 **Nawaz, M. O.**; Henze, D. K. National Climate Action Can Ameliorate, Perpetuate, or Exacerbate International Air Pollution Inequalities. *Nature Communications* **2026**. https://doi.org/10.1038/s41467-026-68827-0.

2025 Siu, T. K.; Goldberg, D. L.; Kerr, G. H.; Chen, L.; **Nawaz, M. O.**; Chang, R. Y.-W.; Fong, K. C. Tropospheric NO2 Patterns in Eastern Canada Using the First Year of TEMPO Observations. *Journal of Geophysical Research: Atmospheres* **2025**, *130* (24), e2025JD044757. https://doi.org/10.1029/2025JD044757.

2025 Goldberg, D. L.; **Nawaz, M. O.**; Lyu, C.; He, J.; Carlton, A. G.; Kondragunta, S.; Anenberg, S. C. Clear-Sky and Cloudy-Sky Differences in NO_2 Concentrations over the United States: Implications for Satellite Measurement Applications. *Atmospheric Chemistry and Physics* **2025**, *25* (22), 16287–16302. https://doi.org/10.5194/acp-25-16287-2025.

2025 **Nawaz, M. O.**; Huber, D. E.; Kerr, G. H.; Judd, L. M.; Acker, S. J.; Goldberg, D. L. A Comparative Analysis of TEMPO NO2 Remote Sensing With Surface-Level Monitoring Through Diurnal and Seasonal Trends, Meteorology, and Monitor Characteristics. *Journal of Geophysical Research: Atmospheres* **2025**, *130* (20), e2025JD043923. https://doi.org/10.1029/2025JD043923.

2025 Hay, S. I.; Ong, K. L.; Santomauro, D. F.; … **Nawaz, M. O.**; … et al. Burden of 375 Diseases and Injuries, Risk-Attributable Burden of 88 Risk Factors, and Healthy Life Expectancy in 204 Countries and Territories, Including 660 Subnational Locations, 1990–2023: A Systematic Analysis for the Global Burden of Disease Study 2023. *The Lancet* **2025**, *406* (10513), 1873–1922. https://doi.org/10.1016/S0140-6736(25)01637-X.

2025 Jin, L.; Benoit, J.; **Nawaz, M. O.**; Rodrigues, P. F.; Wiecko, P.; Miller, J.; Alvarez, G.; Henze, D. K.; Osipova, L.; Anenberg, S. C. Global Health Benefits of Policies to Reduce On-Road Vehicle Pollution through 2040. *Environ. Res. Lett.* **2025**. https://doi.org/10.1088/1748-9326/adcd87.

2025 Wiecko, P.; Henze, D. K.; **Nawaz, M. O.** Sector-, Season-, and Country-Specific NO2-Associated Health Benefits from NOx Emission Reductions. *ACS EST Air* **2025**, *2* (4), 700–709. https://doi.org/10.1021/acsestair.5c00012.

2025 **Nawaz, M. O.**; Goldberg, D. L.; Kerr, G. H.; Anenberg, S. C. TROPOMI Satellite Data Reshape NO2 Air Pollution Land-Use Regression Modeling Capabilities in the United States. *ACS EST Air* **2025**, *2* (2), 187–200. https://doi.org/10.1021/acsestair.4c00153.

2024 Dyer, G. M. C.; Khomenko, S.; Adlakha, D.; Anenberg, S.; Angelova, J.; Behnisch, M.; Boeing, G.; Chen, X.; Cirach, M.; de Hoogh, K.; Diez Roux, A. V.; Esperon-Rodriguez, M.; Flueckiger, B.; Gasparrini, A.; Iungman, T.; Khreis, H.; Kondo, M. C.; Masselot, P.; McDonald, R. I.; Montana, F.; Mitchell, R.; Mueller, N.; **Nawaz, M. O.**; Pereira, E.; Pisoni, E.; Prieto-Curiel, R.; Rezaei, N.; Rybski, D.; Ramasco, J. J.; Schifanella, R.; Shabou, S.; Tatah, L.; Taubenböck, H.; Tonne, C.; Velázquez-Cortés, D.; Woodcock, J.; Zhang, Q.; Nieuwenhuijsen, M. Commentary: A Road Map for Future Data-Driven Urban Planning and Environmental Health Research. *Cities* **2024**, *155*, 105340. https://doi.org/10.1016/j.cities.2024.105340.

2024 Goldberg, D. L.; de Foy, B.; **Nawaz, M. O.**; Johnson, J.; Yarwood, G.; Judd, L. Quantifying NOx Emission Sources in Houston, Texas Using Remote Sensing Aircraft Measurements and Source Apportionment Regression Models. *ACS EST Air* **2024**, *1* (11), 1391–1401. https://doi.org/10.1021/acsestair.4c00097.

2024 Dyer, G. M. C.; Khomenko, S.; Adlakha, D.; Anenberg, S.; Behnisch, M.; Boeing, G.; Esperon-Rodriguez, M.; Gasparrini, A.; Khreis, H.; Kondo, M. C.; Masselot, P.; McDonald, R. I.; Montana, F.; Mitchell, R.; Mueller, N.; **Nawaz, M. O.**; Pisoni, E.; Prieto-Curiel, R.; Rezaei, N.; Taubenböck, H.; Tonne, C.; Velázquez-Cortés, D.; Nieuwenhuijsen, M. Exploring the Nexus of Urban Form, Transport, Environment and Health in Large-Scale Urban Studies: A State-of-the-Art Scoping Review. *Environmental Research* **2024**, *257*, 119324. https://doi.org/10.1016/j.envres.2024.119324.

2024 Choi, J.; Henze, D. K.; **Nawaz, M. O.**; Malley, C. S. Source Attribution of Health Burdens From Ambient PM2.5, O3, and NO2 Exposure for Assessment of South Korean National Emission Control Scenarios by 2050. *GeoHealth* **2024**, *8* (8), e2024GH001042. https://doi.org/10.1029/2024GH001042.

2024 **Nawaz, M. O.**; Johnson, J.; Yarwood, G.; de Foy, B.; Judd, L.; Goldberg, D. L. An Intercomparison of Satellite, Airborne, and Ground-Level Observations with WRF–CAMx Simulations of NO_2 Columns over Houston, Texas, during the September 2021 TRACER-AQ Campaign. *Atmospheric Chemistry and Physics* **2024**, *24* (11), 6719–6741. https://doi.org/10.5194/acp-24-6719-2024.

2023 Gu, Y.; Henze, D. K.; **Nawaz, M. O.**; Wagner, U. J. Response of the Ozone-Related Health Burden in Europe to Changes in Local Anthropogenic Emissions of Ozone Precursors. *Environ. Res. Lett.* **2023**, *18* (11), 114034. https://doi.org/10.1088/1748-9326/ad0167.

2023 **Nawaz, M. O.**; Henze, D. K.; Huneeus, N. J.; Osses, M.; Álamos, N.; Opazo, M. A.; Gallardo, L. Sources of Air Pollution Health Impacts and Co-Benefits of Carbon

Neutrality in Santiago, Chile. *Journal of Geophysical Research: Atmospheres* **2023**, *128* (19), e2023JD038808. https://doi.org/10.1029/2023JD038808.

2023 Jo, D. S.; Nault, B. A.; Tilmes, S.; Gettelman, A.; McCluskey, C. S.; Hodzic, A.; Henze, D. K.; **Nawaz, M. O.**; Fung, K. M.; Jimenez, J. L. Global Health and Climate Effects of Organic Aerosols from Different Sources. *Environ. Sci. Technol.* **2023**, *57* (37), 13793–13807. https://doi.org/10.1021/acs.est.3c02823.

2023 Gu, Y.; Henze, D. K.; **Nawaz, M. O.**; Cao, H.; Wagner, U. J. Sources of PM2.5-Associated Health Risks in Europe and Corresponding Emission-Induced Changes During 2005–2015. *GeoHealth* **2023**, *7* (3), e2022GH000767. https://doi.org/10.1029/2022GH000767.

2023 **Nawaz, M. O.**; Henze, D. K.; Anenberg, S. C.; Ahn, D. Y.; Goldberg, D. L.; Tessum, C. W.; Chafe, Z. A. Sources of Air Pollution-Related Health Impacts and Benefits of Radially Applied Transportation Policies in 14 US Cities. *Front. Sustain. Cities* **2023**, *5*. https://doi.org/10.3389/frsc.2023.1102493.

2023 **Nawaz, M. O.**; Henze, D. K.; Anenberg, S. C.; Braun, C.; Miller, J.; Pronk, E. A Source Apportionment and Emission Scenario Assessment of PM2.5- and O3-Related Health Impacts in G20 Countries. *GeoHealth* **2023**, *7* (1), e2022GH000713. https://doi.org/10.1029/2022GH000713.

2021 **Nawaz, M. O.**; Henze, D. K.; Harkins, C.; Cao, H.; Nault, B.; Jo, D.; Jimenez, J.; Anenberg, S. C.; Goldberg, D. L.; Qu, Z. Impacts of Sectoral, Regional, Species, and Day-Specific Emissions on Air Pollution and Public Health in Washington, DC. *Elementa: Science of the Anthropocene* **2021**, *9* (1), 00043. https://doi.org/10.1525/elementa.2021.00043.

2021 Cao, H.; Henze, D. K.; Cady-Pereira, K.; McDonald, B. C.; Harkins, C.; Sun, K.; Bowman, K. W.; Fu, T.-M.; **Nawaz, M. O.** COVID-19 Lockdowns Afford the First Satellite-Based Confirmation That Vehicles Are an Under-Recognized Source of Urban NH3 Pollution in Los Angeles. *Environ. Sci. Technol. Lett.* **2021**, *9* (1), 3–9. https://doi.org/10.1021/acs.estlett.1c00730.

2021 Nault, B. A.; Jo, D. S.; McDonald, B. C.; Campuzano-Jost, P.; Day, D. A.; Hu, W.; Schroder, J. C.; Allan, J.; Blake, D. R.; Canagaratna, M. R.; Coe, H.; Coggon, M. M.; DeCarlo, P. F.; Diskin, G. S.; Dunmore, R.; Flocke, F.; Fried, A.; Gilman, J. B.; Gkatzelis, G.; Hamilton, J. F.; Hanisco, T. F.; Hayes, P. L.; Henze, D. K.; Hodzic, A.; Hopkins, J.; Hu, M.; Huey, L. G.; Jobson, B. T.; Kuster, W. C.; Lewis, A.; Li, M.; Liao, J.; **Nawaz, M. O.**; Pollack, I. B.; Peischl, J.; Rappenglück, B.; Reeves, C. E.; Richter, D.; Roberts, J. M.; Ryerson, T. B.; Shao, M.; Sommers, J. M.; Walega, J.; Warneke, C.; Weibring, P.; Wolfe, G. M.; Young, D. E.; Yuan, B.; Zhang, Q.; de Gouw, J. A.; Jimenez, J. L. Secondary Organic Aerosols from Anthropogenic Volatile Organic Compounds Contribute Substantially to Air Pollution Mortality. *Atmospheric Chemistry and Physics* **2021**, *21* (14), 11201–11224. https://doi.org/10.5194/acp-21-11201-2021.

2021 Malley, C. S.; Hicks, W. K.; Kuylenstierna, J. C. I.; Michalopoulou, E.; Molotoks, A.; Slater, J.; Heaps, C. G.; Ulloa, S.; Veysey, J.; Shindell, D. T.; Henze, D. K.; **Nawaz, M. O.**; Anenberg, S. C.; Mantlana, B.; Robinson, T. P. Integrated Assessment of Global Climate, Air Pollution, and Dietary, Malnutrition and Obesity Health Impacts

of Food Production and Consumption between 2014 and 2018. *Environ. Res. Commun.* **2021**, *3* (7), 075001. https://doi.org/10.1088/2515-7620/ac0af9.

2020 **Nawaz, M. O.**; Henze, D. K. Premature Deaths in Brazil Associated With Long-Term Exposure to PM2.5 From Amazon Fires Between 2016 and 2019. *GeoHealth* **2020**, *4* (8), e2020GH000268. https://doi.org/10.1029/2020GH000268.

Manuscripts Under Review (9)

2026 Lang, V.A., Partha, D., Goldberg, D.L., **Nawaz, M.O.**, Kerr, G., Chen, L., Janssen, M., Adelman, Z.E., Urbaszewski, B., Deylami, N., Minuche, M., Chavez, G., Horton, D.E. A framework for augmenting medium- and heavy-duty vehicle emissions within the National Emissions Inventory for hyperlocal air quality assessments.

2026 Hu, F., Zhuo, L., **Nawaz, M.O.**, Wen, Z., Han, D. Improving TROPOMI Methane Plume Candidate Screening: Gradient Boosting, Chemical Context, and Image-Based Approaches.

2026 Graffy, P.M., … **Nawaz, M.O.**, … et al. The association of air pollution with death and deterioration on the United States transplant waiting list.

2026 Kim, S.Y., Kerr, G.H., **Nawaz, M.O.**, Anenberg, S.C. Fine-scale spatiotemporal patterns of NO2 pollution and associated mortality burdens across the continental United States.

2026 Gu, Y., Henze, D.K., **Nawaz, M.O.**, Zhang, L., Liao, H. Non-agricultural NH_3 emissions disproportionately contribute to urban $PM_{2.5}$ health burdens in China.

2026 Graffy, P.M., … Nawaz, **M.O.**, … et al. Respiratory Trajectory Phenotypes and Air Pollution-Associated Outcomes in Critically Ill Patients with Lung Cancer in the United States.

2026 Graffy, P.M., … Nawaz, **M.O.**, … et al. Cumulative Air Pollution Exposure Increases the Severity of Acute Respiratory Failure: a U.S. Multicenter Study.

2025 Morris, S. T., O'Neill, B. C., Msangi, S., **Nawaz, M.O.**, Parker, N., Rao, N., Van Vuuren, D., et al. Modeling the human well-being dimensions of global change: priorities and challenges for research to inform decision-making.

2025 Kerr, G.H., **Nawaz, M.O.**, Anenberg, S.C., Anthoff, D., Burton, C., Carter, T.S., Henze, D.K., Kelley, D.I., Kingdon, C., O'Dell, K., Prest, B.C., Cromar, K.R. Climate-driven surges in public health damages from wildland fire-sourced pollution through 2100.

RESEARCH SUPERVISION

Postdoctoral

2026- Lucas Ngo, Cardiff University (**Primary Supervisor**)

PhD Supervision

2026- Diya Wang, Cardiff University (**Primary Supervisor**)

2025- Feichi Hu, Cardiff University (**Co-Supervisor** with L. Zhuo as Primary)

PhD Examination

2026- Xin Wang, Cardiff University (**Internal Review Chair**)

2026- Patrick Wiecko, CU Boulder (**External Review Panel Member**)

MSc, MSci and MPH

2026- Mia Kate Petts, Cardiff University (**Primary Supervisor**)
2026- Mohammed Fahad H Alsuwayh, Cardiff University (**Primary Supervisor**)
2025-2026 Kyla Camama (MSci), Cardiff University (**Primary Supervisor**)
2023-2024 Katie O'Donnell, GWU (**Co-Supervisor** with S. Anenberg as Primary)
2023-2024 Erin Campbell, GWU (**Co-Supervisor** with S. Anenberg as Primary)

BSc

2025- 10 BSc Geography Dissertations, Cardiff University (**Primary Supervisor**)
2024-2025 Olivia Paquette, GWU (**Co-Supervisor** with D. Goldberg as Primary)
2019-2021 Mohammed Alwakeel, CU Boulder (**Co-Supervisor** with D. Henze as Primary)

TEACHING

Cardiff University, Department of Earth and Environmental Sciences

Title	Level	Role
The Ocean-Atmosphere System	Year 2	Module Leader (2025-)
GIS, Maps, and Analytical Skills	Year 1	Lecturer (2025-)
Climate Change Adaptation and Resilience	MSc	Co-teacher (2026-)
Digital Fieldwork Workshop	Year 1	Lecturer (2025-)
Geography Tutorials	Year 1/2/3	Personal Tutor (2025-)
BSc Dissertations	Year 3	Supervisor (2025-)
MSc Dissertations	Postgraduate	Supervisor (2025-)

George Washington University, Department of Environmental and Occupational Health

Title	Level	Role
Global Climate Change & Air Pollution	MPH	Lecturer (2024)

University of Colorado Boulder, Department of Mechanical Engineering

Title	Level	Role
PhD Preliminary Exam Review	PhD	Mentor (2020)
Numerical Methods for Engineers	Year 1	Lead TA (2018-2019)

PROFESSIONAL ASSOCIATIONS

2025- EQUATOR Mentor
2025- TOAR-II Health Team
2025- COST Action CliMent Member (CA23113)
2024- Global Burden of Disease Study Collaborator
2023- European Geosciences Union
2018- American Geophysical Union

SELECTED MEDIA COVERAGE

BBC News Radio Wales, *Cleaner air in steel town a year since furnace shut*, https://www.bbc.co.uk/sounds/play/p0mq8hsb, 23 December 2025

WXPR, *The health impacts of tailpipe emissions get a fresh look*, https://www.wxpr.org/energy-environment/2025-08-11/the-health-impacts-of-tailpipe-emissions-get-a-fresh-look, 11 August 2025

Eos Earth and Space Science News, *The Global Health Benefits of Going Net Zero*, https://eos.org/research-spotlights/the-global-health-benefits-of-going-net-zero, 14 February 2023

Revista Pesquisa FAPESP, *Queimadas na Amazônia aumentam internações* [*Fires in the Amazon increase hospitalizations*], https://revistapesquisa.fapesp.br/queimadas-na-amazonia-aumentam-internacoes/, 18 September 2020

DEPARTMENTAL SERVICE

Cardiff University, School of Physical, Chemical and Environmental Sciences

2026- Equality, Diversity, Inclusion & Anti-Racism Lead
This is my primary administrative role in which I lead EDIA strategy and culture, advise on equality and inclusion matters, and oversee delivery of chartermark activity
2026 Member of Future Generations Engagement Fund Review Panel
2026 Member of Staff Promotions Panel
2026 Member of Extenuating Circumstances Panel
2025 Director of EDI (for former School of Earth and Environmental Sciences)
2025- Member of Research Ethics Committee
2025 Member of Research Leave Panel
2025 Organiser of School EDI Away Day
2025- Contributor to Induction for Year 1 Students
2025 Open Day Staff Participant

ACADEMIC SERVICE

Ad Hoc Peer Review for Journals (46)

Atmospheric Chemistry and Physics (1); Communications Earth & Environment (2); Discover Atmospheres (1); Discover Cities (1); Earth's Future (1); Earth System Science Data (1); Elementa: Science of the Anthropocene (1); Environmental Monitoring & Assessment (1); Environmental Research Letters (2); Environmental Science & Technology (3); Environmental Science & Technology Air (1); Environmental Science Policy Research (1); GeoHealth (3); Geoscience Letters (1); Health Data Science (1); Humanities & Social Science Communications (1); International Geoscience and Remote Sensing Symposium (6); Journal of the Air and Waste Management Association (1); Lancet Planetary Health (1); Nature (1); Nature Aging (1); Nature Cities (1); Nature Communications (1); Nature Food (1); Nature Health (4); Nature Sustainability (1); NPJ Clean Air (1); NPJ Climate & Atmospheric Science (1); Open Research Europe (1); PLOS (1); PNAS (1); Scientific Reports (1)

Ad Hoc Peer Review for Proposals (2)

Wellcome Trust Discretionary Award Expert Review (2)

Editorial Roles

2026- Associate Editor for *Air Quality, Atmosphere & Health*
2025 Associate Editor for *Frontiers in Environmental Health* (Environmental Epidemiology)

Other Professional Service

2026 Session convener on air quality, climate, health, and equity for EGU 2026 meeting
2025 Session convener on air quality, climate, health, and equity for EGU 2025 meeting

PRESENTATIONS (31)

Invited Talks (4)

2026 The added benefit of TEMPO remote sensing of NO_2 for exposure and health impact assessment. June 1st-5th. CEOS Atmospheric Composition-Virtual Constellation. Leiden, Netherlands.

2026 Using HPC for the variable resolution simulation of air pollution formation using the GCHP stretched-grid model. May 20th. Advanced Research Computing Operational & User Group meeting. Cardiff, Wales, UK.

2025 Applying machine learning and statistical modeling approaches to remote sensing observations for inferring surface-level NO_2. December 15th-19th. American Geophysical Union. New Orleans, LA, USA.

2023 Using satellite data to characterize air pollution and health in cities and countries. April 27th. Environmental Defense Fund / Climate and Clean Air Coalition / United Nations Environment Programme Workshop for Clean Air Solutions in Latin America and the Caribbean. Bogotá, Colombia.

Oral Presentations (12)

2025 **Nawaz, M.O.**, Henze, D.K., Exploring the role of climate action in transboundary air pollution inequality using GEOS-Chem adjoint sensitivities. April 29th. European Geosciences Union. Vienna, Austria.

2024 **Nawaz, M.O.**, Anenberg, S.C., Goldberg, D.L., Kerr, G.H., Kondragunta, S. Development of a Land-Use Regression of Hourly Surface NO2 in preparation for GeoXO Atmospheric Composition Data. April 17th. European Geosciences Union. Vienna, Austria.

2024 **Nawaz, M.O.**, O'Dell, K., Anenberg, S.C., Goldberg, D.L., Kerr, G.H., He, J., McDonald, B., Kondragunta, S. Value of GeoXO Atmospheric Composition Data for Estimating Air Pollution-Related Health Impacts. January 30th. American Meteorological Society. Baltimore, MD, USA.

2023 **Nawaz, M.O.,** Henze, D.K., Anenberg, S.C., Goldberg, D.L., Investigating climate co-benefits using GEOS-Chem adjoint sensitivities. August 15th. Second GEOS-Chem Europe Meeting. London, UK.

2023 **Nawaz, M.O.,** Henze, D.K., Anenberg, S.C., Tessum, C. Regional vs local sources of municipal air pollution-related health impacts. January 10th. American Meteorological Society. (Presented by Henze). Denver, CO, USA.

2022 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Huang, T. Developing an interactive tool for characterizing the air pollution-related health impacts in Los Angeles, CA associated with different proposed emission scenarios. July 19th. Earth Science Information Partners Meeting. Pittsburgh, PA, USA (Virtual).

2022 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Harkins, C., Gallardo, L., Barazza Basoa, K. Leveraging satellite-derived data in GEOS-Chem adjoint simulations to characterize the sources of PM2.5-, O3-, and NO2-related health impacts at multiple spatial scales. June 9th. 10th International GEOS-Chem Meeting. St. Louis, MO, USA. (Virtual).

2022 **Nawaz, M.O.**, Henze, D.K., Braun, C., Miller, J., Pronk, E., Anenberg, S.C. Characterizing the sources of air pollution at the urban- and country-scale: case studies in Santiago, Chile and G20 countries. February 17th. Graduate Engineering Annual Research and Recruitment Symposium. Boulder, CO, USA.

2021 **Nawaz, M.O.**, D. Henze, S.C. Anenberg, C. Braun, J. Miller. Comparing domestic and extra-regional contributions to pollutant exposures and health impacts in G20 countries through a novel adjoint modeling approach. December 15th. American Geophysical Union Fall Meeting. New Orleans, LA, USA (Virtual).

2020 **Nawaz, M.O**., D. Henze, D. Goldberg, S. Anenberg, D. Jo, B. Nault, J.L. Jimenez, H. Cao, C. Harkins, Z. Qu. Characterizing the regional, sectoral and species-specific sources of pollution exposure and its associated health impacts in urban environments: case studies in Washington, D.C. and Santiago, Chile. December 14th. American Geophysical Union. (Virtual)

2020 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Goldberg, D. Premature deaths in Brazil associated with long-term exposure to PM2.5 from Amazon fires and development of a nested South American domain for the GEOS-Chem Adjoint. June 23rd. 19th GEIA Conference. (Virtual)

2019 **Nawaz, M.O.**, Henze, D.K. Source attribution of PM2.5 from sensitivity analyses in the GEOS-Chem adjoint model. October 25th. Young Scientists Symposium on Atmospheric Research. Fort Collins, CO, USA.

Poster Presentations (15)

2026 **Nawaz, M.O.**, Goldberg, D.L., Anenberg, S.C., Kerr, G.H. Estimating the air pollutant-attributable health burden of the oil and gas sector in Mexico using a TROPOMI flux-divergence approach. May 7th. European Geosciences Union. Vienna, Austria.

2025 **Nawaz, M.O.**, Goldberg, D.L., Anenberg, S.C., Kerr, G.H. What does low-earth orbiting, geostationary, and airborne remote-sensing reveal about surface NO_2? September 9th. UK Atmospheric Chemistry Conference. York, UK.

2025 **Nawaz, M.O.**, Southerland, V.A., Goldberg, D.L. Characterizing the air quality and health impacts from oil and gas emissions in Mexico using GCHP. August 19th. GEOS-Chem Europe Meeting 3. London, UK.

2024 **Nawaz, M.O.,** Goldberg, D.L., Kerr, G.H., Anenberg, S.C., What can TROPOMI and TEMPO remote sensing reveal about seasonal and diurnal trends in surface-level NO_2? December 13th. American Geophysical Union. Washington, DC, USA.

2022 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Harkins, C., Gallardo, L., Barazza Basoa, K. Leveraging satellite-derived data and air quality modeling to characterize source profiles of climate co-benefits at the urban- and country- scale. December 12th. American Geophysical Union. Chicago, IL, USA.

2020 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, A. Van Donkelaar, R. Martin, M. L. Serre, J. J. West. Health benefits of decreases in PM2.5 and ozone in the United States, 1990-2016. July 21st. NASA Health and Air Quality Applied Sciences Team Final Showcase. (Virtual).

2019 **Nawaz, M.O.**, D.K. Henze, S.C. Anenberg, D. Goldberg, Z. Qu. Source attribution of PM2.5 and O3 concentrations and health outcomes from 2010 and 2011 in Washington D.C. using sensitivity analyses in the GEOS-Chem adjoint model. December 9th. American Geophysical Union, San Francisco, CA, USA

2019 **Nawaz, M.O.**, D.K. Henze, C.S. Malley, J.C.I. Kuylenstierna, H.W. Vallack, Y. Davila, S.C. Anenberg, S. Terry, A. Curry-Brown, N. Fann, E. Lefevre, C. Heaps, S. Penn, H. Roman, J. Neumann. Source attribution of climate and health impacts from aerosols. May 6th. 9th International GEOS-Chem Meeting, Cambridge, MA, USA.

2019 **Nawaz, M.O.**, Henze, D.K., The use of adjoint modeling to assess the sources of air pollution and its associated health impacts. February 21st. Graduate Engineering Annual Research and Recruitment Symposium. Boulder, CO, USA.

2018 **Nawaz**, **M.O.**, D. K. Henze, C. Malley, GH41C-1446: Source Attribution of Climate and Health Impacts from Aerosols. December 14th. AGU Fall Meeting. Washington, DC, USA.

2018 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, A. van Donkelaar, R. V. Martin, J. J. West. Health benefits of decreases in PM2.5 and ozone in the United States, 1990-2015. July 16th. NASA Health and Air Quality Applied Sciences Team Meeting. Madison, WI, USA.

2018 **Nawaz, M.O.,** Y. Zhang, D. Q. Tong, A. van Donkelaar, R. V. Martin, J. J. West. Health benefits of decreases in PM2.5 and ozone in the United States, 1990-2015. April 20th. Climate Change and Resilience Symposium. Chapel Hill, NC, USA.

2017 **Nawaz, M.O.,** Y. Zhang, D. Q. Tong, J. J. West. Health benefits of decreases in PM2.5 and ozone in the United States from 1990 to 2015. December 11th. American Geophysical Union. New Orleans, LA, USA.

2017 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, J. J. West. Health benefits of decreases in PM2.5 and ozone in the United States from 1990 to 2015. October 23rd. Community Modeling and Analysis System Conference. Chapel Hill, NC, USA.

2017 **Nawaz, M.O.**, Y. Zhang, West, J.J. Impact of regional ozone precursor emissions on global ozone burden. April 12th. Celebration of Undergraduate Research. Chapel Hill, NC, USA.